

# Mukundan, Malavika

<https://MalavikaMukundan.github.io>

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|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| EMPLOYMENT        | Postdoctoral Associate<br>Department of Mathematics and Statistics<br>Boston University, Boston, MA<br>Mentor: David Fried                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sep 2024 - Present     |
|                   | President's Postdoctoral Fellow<br>Department of Mathematics<br>University of California, Los Angeles, CA<br>Mentor: Mario Bonk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Jul 2024 - Aug 2024    |
| EDUCATION         | University of Michigan, Ann Arbor, MI<br>PhD in Mathematics<br>Advisor: Sarah C. Koch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sep 2018 - May 2024    |
|                   | MS in Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sep 2018 - Aug 2020    |
|                   | Chennai Mathematical Institute, Chennai, India<br>BSc. (Hons.) in Mathematics and Computer Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jul 2015 - Apr 2018    |
|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                        |
| PAPERS            | Published and Submitted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                        |
|                   | <ul style="list-style-type: none"><li>Dynamical approximations of post-singularly finite entire functions (joint with N. Prochorov and B. Reinke)<br/>Accepted at <i>Transactions of the AMS</i><br/>arXiv:2305.17793</li><li>Donnay et al., (2025). VoteKit: A Python package for computational social choice research.<br/><i>Journal of Open Source Software</i>, 10(109), 7477<br/><a href="https://joss.theoj.org/papers/10.21105/joss.07477">https://joss.theoj.org/papers/10.21105/joss.07477</a></li><li>Embedding unicritical connectedness loci<br/><i>Conform. Geom. Dyn.</i> 28 (2024), 131-164<br/><a href="https://doi.org/10.1090/ecgd/389">https://doi.org/10.1090/ecgd/389</a></li><li>A solution to the degree-<math>d</math> twisted rabbit problem (joint with R. Winarski)<br/><i>Ergodic Theory and Dynamical Systems</i>. Published online 2025:1-16.<br/>doi:10.1017/etds.2025.137</li><li>A cell decomposition for marked cycle curves (joint with C. Davis, D. Stoll and G. Tiozzo)<br/>Submitted; arXiv:2410.21049</li></ul> |                        |
|                   | Preprints                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                        |
|                   | <ul style="list-style-type: none"><li>Dynamical approximation of post-singularly finite exponentials<br/>arXiv:2305.18245</li><li>Candidate Moderation under Instant Runoff and Condorcet Voting: Evidence from the Cooperative Election Study (joint with D. McCune, M. I. Jones, A. Schultz, A. Graham-Squire, I. Volic, B. See, K. Xiao)<br/>arXiv:2603.03619</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                        |
| GRANTS<br>&AWARDS | AMS-Simons Travel Grant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jul 2025 - Jun 2027    |
|                   | AMS Graduate Student Travel Grant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Feb 2024               |
|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                        |
|                   | University of Michigan, Ann Arbor, MI<br>Rackham Conference Travel Grant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Jun 2023, Dec 2023     |
|                   | Usha Sharma Bhalla Fellowship Fund                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Summer 2022, Fall 2023 |

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|-----------------------------------------|-------------------------|
| Carroll V. Newsom Scholarship           | May 2023                |
| Rackham Graduate Student Research Grant | 2023                    |
| Departmental Fellowship                 | Summer 2020, 2021, 2022 |

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| Lauter Program Associate Fellowship, MSRI, Berkeley, CA | Jan - May 2022 |
| Stipend, Chennai Mathematical Institute, Chennai, India | 2015 - 2018    |
| Summer Research Fellowship, JNCASR, Bengaluru, India    | May - Jun 2017 |

## INVITED TALKS

### Conferences

*59th Annual Spring Topology and Dynamical Systems Conference*, Birmingham, Alabama, Mar 2026

*Joint Mathematics Meetings: Special Session on the Mathematics of Decisions, Elections and Games*, Washington D.C., Jan 2026

*AMS Fall Eastern Sectional Meeting*, St Louis University, MO, Oct 2025

*Social Choice: Theory and Computation, An Interdisciplinary Conference on Voting, Representation, and Districting*, Institute for Mathematics and Democracy, Wellesley College, MA, Oct 2025

*DKO Workshop and Summer School on Transcendental Thurston Theory*, The University of Manchester, United Kingdom, Aug 2025

*JHH80 Dynamical Developments: Degenerations of Flat Surfaces and Rational Maps*, Institut de Mathématiques de Toulouse, France, Jun 2025

*Topics in Complex Dynamics 2025: Connections to Geometry and Analysis*, Universitat de Barcelona, Spain, Jun 2025

*Degeneration Techniques for Rational Maps*, Warwick Mathematics Institute, University of Warwick, United Kingdom, May 2025

*Early Career Complex Dynamics Workshop*, Indiana University Bloomington, IN, Jul 2024

*Midwest Dynamical Systems Early Career Conference*, The Ohio State University, OH, May 2024

*17th Annual Bloomington Geometry Workshop*, Indiana University Bloomington, IN, Apr 2024

*Joint Mathematics Meetings: Special Session on the Mathematics of Decisions, Elections and Games*, San Francisco, CA, Jan 2024

*AMS Fall Southeastern Sectional Meeting: Special Session on Ergodic Theory and Dynamical Systems*, University of South Alabama, AL, Oct 2023

*Early Career Complex Dynamics Workshop*, Indiana University Bloomington, IN, Jun 2023

*CMS Summer Meeting: Session on Geometric Topology, pseudo-Anosov Maps and Complex Dynamics*, Ottawa, Canada, Jul 2023

### Colloquia & Seminars

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|-----------------------------------------------|----------|
| <i>College of William and Mary Colloquium</i> | Feb 2026 |
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| <i>Tufts University Dynamical Systems and Ergodic Theory Seminar</i> | Dec 2025 |
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| <i>University of South Alabama Colloquium</i> | Nov 2025 |
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| <i>Harvard Informal Seminar: Geometry, Dynamics and Moduli Spaces</i> | Sep 2025 |
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| <i>Chennai Mathematical Institute Colloquium</i> | Aug 2025 |
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| <i>Tata Institute of Fundamental Research Lecture</i> | Jul 2025 |
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| <i>New England Dynamics Seminar</i> | Apr 2025 |
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| <i>Brandeis University Everytopic Seminar</i> | Apr 2025 |
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| <i>Harvard University Algebraic Dynamics Seminar</i> | Oct 2024 |
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| <i>Boston University Dynamical Systems Seminar</i> | Jan, Sep 2024 |
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| <i>IMS and SUNY Stony Brook Dynamics Seminar</i> | Aug 2024 |
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| <i>University of Illinois Chicago Dynamics Seminar</i> | Jan 2024 |
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| <i>Quasiworld Seminar (online)</i> | Dec 2023 |
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| <i>Cornell University Dynamics Seminar</i> | Nov 2023 |
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| <i>University of Michigan CADG Seminar</i> | Nov 2020, Apr, Oct 2021, Nov 2023 |
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| <i>The Open University Dynamical Systems Seminar (online)</i> | Apr 2023 |
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| <i>University of Liverpool Dynamical Systems Seminar</i> | Mar 2023 |
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| <i>University of Warwick</i> Geometry and Topology Seminar              | Mar 2023 |
| <i>Institute of Mathematical Sciences</i> Chennai, Colloquium           | Dec 2022 |
| <i>University of Toronto</i> Dynamics Seminar                           | Nov 2022 |
| <i>University of Chicago</i> Topology and Geometry Seminar              | Oct 2022 |
| <i>TIFR Infosys Chandrasekharan</i> Random Geometry Colloquium (online) | Oct 2022 |
| <i>MSRI</i> Complex Dynamics Junior Seminar                             | Apr 2022 |
| <i>IUPUI</i> Dynamical Systems Seminar                                  | Nov 2021 |

## TEACHING

|                                                    |             |
|----------------------------------------------------|-------------|
| <b>Boston University</b> , Boston, MA              |             |
| <i>MA 564</i> (Introduction to Topology)           | Spring 2026 |
| <i>MA 871</i> (Dynamical Systems Graduate Seminar) | Fall 2025   |
| <i>MA 771</i> (Introduction to Dynamical Systems)  | Spring 2025 |
| <i>MA 123</i> (Calculus I)                         | Fall 2024   |

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| <b>University of Michigan</b> , Ann Arbor, MI     |                                          |
| <i>Math 116</i> (Calculus II)                     | Fall 2021, Winter 2020, 2021, 2023, 2024 |
| <i>Math 105</i> (Data, Functions and Graphs)      | Fall 2020                                |
| <i>Math 215</i> (Calculus III) Teaching Assistant | Fall 2019                                |
| <i>Math 115</i> (Calculus I)                      | Fall 2018, Winter 2019                   |

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| <b>Chennai Mathematical Institute</b> , Chennai, India |             |
| <i>Analysis II</i> Teaching Assistant                  | Spring 2018 |
| <i>Analysis I</i> Teaching Assistant                   | Fall 2017   |

## SERVICE

### Refereeing

Transactions of the AMS  
 Journal für die reine und angewandte Mathematik (Crelle's Journal)  
 Mathematische Annalen  
 Discrete and Continuous Dynamical Systems

### Organizing

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| The Baby Mandelbrot Set Meetings (virtual)                              | 2025 - Present |
| Boston University Dynamical Systems Seminar                             | 2024 - Present |
| <i>DKO Workshop and Summer School on Transcendental Thurston Theory</i> |                |
| University of Manchester, United Kingdom                                | Aug 2025       |
| UM Jobs in Academia for Mathematics (JAMS) Sessions                     | Apr 2023       |
| <i>Michigan Research Experience for Graduates</i> , Ann Arbor, MI       | Jun 2022       |

### Mentoring

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| AWM Mentor                                                       | 2023 - Present |
| <i>Bachelor's Thesis Committee Member</i> for Hunter Golemon, BU | Spring 2025    |
| <i>Teaching Assistant</i> , BIG Ideas in Dynamics                | Fall 2024      |

### Miscellaneous

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| <i>Reviewer</i> , MathSciNet                                    | Dec 2025 - Present  |
| <i>Co-Chair</i> , UM Graduate Student Advisory Committee        | Sep 2022 - May 2024 |
| <i>Grad Member</i> , UM Math Department Climate Committee       | Sep 2022 - Aug 2023 |
| <i>Volunteer Artist</i> , AWM EvenQuads Deck 2                  | 2023                |
| <i>Board Member</i> , UM AWM Student Chapter                    | Sep 2020 - Aug 2022 |
| <i>Steward</i> , UM Graduate Employees' Organization (GEO) 3550 | Sep 2021 - Aug 2022 |
| <i>Volunteer</i> , Michigan Math Corps summer camp              | Jul 2020            |

## CONFERENCES & WORKSHOPS

*Arithmetic, Algebraic and Analytics Dynamics*, CIRM, Luminy, France, Jan 2025  
*Conference on Holomorphic Dynamics and Fractal Geometry*, BICMR, China, Jan 2025  
*Thematic Program on Randomness and Geometry*, Fields Institute, Toronto, Canada,

Apr 2024

*Randomization, Neutrality, and Fairness*, Simons Laufer Mathematics Institute, Berkeley, CA, Oct 2023

*Quasiworld Workshop*, University of Helsinki, Finland, Aug 2023

*Around the Mandelbrot set: A conference celebrating the 60th birthday of Mitsuhiro Shishikura* (online participation), Tokyo, Japan, May 2023

*Midwest Dynamical Systems Early Career Conference*, University of Notre Dame, IN, May 2022

*Complex Dynamics School: Building a Bridge to Higher Dimensions*, IUPUI, IN, Mar 2022

*Complex Dynamics: from Special Families to Natural Generalizations in One and Several Variables*, MSRI, Berkeley, CA, Jan - May 2022

*Midwest Dynamical Systems Conference*, Northwestern University, IL, Nov 2021

*Advancing Bridges in Complex Dynamics* (online participation), CIRM, Luminy, France, Sep 2021

*Midwest Dynamical Systems Conference*, University of Illinois Chicago, IL, Nov 2019

*Summer School on Dynamical Systems*, University of Houston, TX, Jun 2019

*School on Low-Dimensional Geometry and Topology: Discrete and Algorithmic Aspects*, Institut Henri Poincaré, France, Jun 2018

## COMPUTER SKILLS

Python

Julia

C++

Haskell

Git